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Driving open innovation in SMEs: the role of organizational and strategic dynamics

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Abstract

The primary focus of this research is on Small and Medium-Sized Enterprises (SMEs) in Singapore engaged in manufacturing activities, particularly those involved in the production of technology products. Although numerous studies have explored open innovation, further research is essential to investigate the interconnected factors influencing its adoption. This study aims to examine the relationship between organizational citizenship behaviours, integrative culture, management linkages, transactional costs, and the role of appropriability regimes in shaping SMEs' inclination to adopt open innovation. Adopting a positivist paradigm, this research employs quantitative methodologies with a deductive research approach. Hypotheses were developed to explore the impact of the identified factors. Primary data were collected through a survey targeting SMEs, utilizing a Likert-scale-based questionnaire. Confirmatory Factor Analysis (CFA) was conducted to evaluate the factor structure and validate the measurement model. Subsequently, Partial Least Squares Structural Equation Modelling (PLS-SEM) was employed to assess the relationships between variables and ensure the internal consistency of the derived factors. The findings highlight the potential of open innovation practices in SMEs to enhance technological outcomes, resulting in the development of advanced, high-quality, and ecologically sustainable products and services. These practices contribute significantly to societal well-being and sustainability. Furthermore, the integration of technological innovation fosters the educational development of future management leaders, equipping them with essential skills for managing innovation in a globalized world. Establishing collaborative relationships across diverse industries promotes organizational learning and cultivates a proactive approach to innovation management, addressing critical challenges in the era of globalization.

Keywords: Open innovation, Organizational citizenship behaviour, SMEs, Integrative culture, Managerial ties, Transaction cost, Manufacturing, Appropriability regime, Sustainability

Introduction

Open innovation, a paradigm that leverages both internal and external information flows to drive innovation and market expansion, has gained considerable attention in recent years (Pedota & Piscitello, 2022). It allows businesses to engage with external networks to co-create and distribute innovations, fostering a more dynamic approach to innovation management. Sharing knowledge, gaining new perspectives and incorporating external feedback are fundamental elements of this strategy as this framework has proven essential for enabling informed decision making, as more significant knowledge acquisition enhances the quality of strategic choices (Cheng & Shiu, 2021). However, the implementation of open innovation strategies varies widely, particularly across the Asia–Pacific region, with mixed success rates. To bolster innovation capabilities, businesses must continuously refine and adapt their open innovation practices. In the Asia–Pacific region, open innovation strategies impact the outcomes of innovation processes differently, influenced by the complexity of these processes and their interconnectedness (Remneland Wikhamn & Styhre, 2023). Effective collaboration in knowledge-intensive domains is critical for navigating elevated process dependencies and complexities. Southeast Asia has experienced an accelerated shift towards digitalization and exacerbated by the COVID-19 pandemic and this rapid transition has exposed significant vulnerabilities in organizations reliant on outdated technologies, highlighting the necessity for innovative approaches (Sarwar et al., 2023). Open innovation has emerged as a pivotal strategy for businesses in the Asia–Pacific region to achieve agility, adaptability and customer-centricity in an increasingly competitive landscape. Small and medium-sized enterprises (SMEs) face unique challenges such as resource constraints in adopting open innovation as a mechanism for overcoming technological and market-related hurdles. Singapore has been recognized as a gateway to the Asia–Pacific region and exemplifies a thriving innovation hub within Southeast Asia as its strategic location and infrastructure foster knowledge exchange and innovation collaboration, enabling regional and global innovation corridors (Zhang & Khanal, 2024). Digital innovation in Asia, underpinned by robust ecosystems like Singapore is projected to play a leading role in global open innovation paradigms. This interconnected ecosystem facilitates cross-border collaboration and empowers individuals and organizations to develop transformative solutions that drive economic growth and societal advancement.

The role of SMEs in Singapore's innovation landscape

SMEs are integral to Singapore's economy contributing significantly to its manufacturing and service sectors, but they often lag behind their global counterparts in technology adoption and innovation (ADB, 2022). Digitalization is increasingly viewed as a competitive necessity but many SMEs in Singapore continue to rely on manual processes and traditional practices and this reliance poses significant barriers to scaling operations globally and fully integrating into the open innovation ecosystem as intellectual property concerns limited access to external networks and insufficient internal competencies exacerbate these challenges (Ragazou et al., 2022). Furthermore, the resource-constrained environments typical of SMEs restrict their ability to effectively identify and leverage external opportunities. Notably, there is a dearth of scholarly investigations into the adoption of open innovation among SMEs in the Asia–Pacific region, particularly in Singapore as existing research fails to adequately address the nuances of open innovation practices within SMEs and their implications for management strategies (Pérez-Orozco et al., 2024). This study aims to fill this gap by examining how SMEs in Singapore can leverage open innovation to overcome resource constraints and enhance global competitiveness. Given the Singaporean government's commitment to enhancing SME productivity and technical proficiency in the post-COVID-19 era, this research provides timely insights into fostering sustainable growth through open innovation.

Challenges and opportunities for open innovation in SMEs

SMEs in Singapore face numerous challenges in adopting open innovation practices including limited financial resources inadequate internal capabilities and restricted access to external networks (Rangaswamy et al., 2024). While Singapore's reputation as a technological hub is well-established, the innovation performance of its SMEs often falls short compared to peers in other high-income countries (Cheang, 2024). Addressing these challenges necessitates integrating open innovation principles into business frameworks to unlock growth opportunities and enhance value creation. This integration is particularly critical given the increasing reliance on digital technologies in the global business landscape. Employee retention presents another significant challenge for SMEs (Biea et al., 2024). High-performing individuals often leave due to dissatisfaction, excessive workloads, or limited career advancement opportunities, particularly when management practices fail to create supportive environments. Strategic management practices, coupled with fostering organizational citizenship behaviour (OCB), are crucial for retaining talent and driving organizational success (Halid et al., 2024). OCB, characterized by informal, hierarchical, and personal interaction patterns, plays a distinctive role in SMEs. It encourages employees to exceed expectations, enhancing team performance and overall organizational effectiveness. Understanding the interplay of managerial practices and employee behaviours is vital for identifying opportunities to improve productivity and innovation capacity.

Theoretical underpinnings and research gap

The theoretical foundation for this study draws from Actor Network Theory (ANT) and Social Exchange Theory (SET). ANT provides a lens for understanding the interconnected networks that facilitate innovation, emphasizing the importance of relationships and the roles of various actors within these networks (Wynn & Garwood-Cross, 2024). SET highlights the reciprocity and mutual benefits inherent in collaborative relationships, aligning with the principles of open innovation (Fang et al., 2024). These frameworks offer valuable insights into how SMEs can navigate the complexities of resource-constrained environments and leverage external networks for innovation. Despite the growing interest in open innovation the literature reveals a significant research gap concerning its application in SMEs particularly within Singapore (Rangaswamy et al., 2024). While previous studies have explored open innovation in large enterprises, the unique challenges and opportunities faced by SMEs remain underexplored. In addition, limited attention has been given to the role of managerial ties, integrative culture, and organizational behaviours in fostering innovation within SMEs (Le & Ha, 2024). This study seeks to address these gaps by examining the interplay of open innovation, managerial practices, and organizational culture in the SME context.

Contributions of the study

This research contributes to the academic and practitioner communities by addressing the research gap, providing empirical insights into open innovation practices among

SMEs in Singapore, and thereby enriching the existing literature. It advances theoretical understanding by applying ANT and SET to the SME context, offering a novel perspective on innovation networks and collaborative practices. The study identifies actionable strategies for SMEs to overcome resource constraints and enhance global competitiveness through open innovation. It also informs policymakers on the support mechanisms required to foster a robust SME innovation ecosystem in Singapore. By highlighting successful open innovation practices among Singaporean SMEs, the research sets benchmarks for others and provides a comparative view of open innovation adoption across different regions to contextualize the unique challenges and opportunities in Singapore.

Practical toolkit for SMEs

To aid SMEs in adopting open innovation, this study proposes a practical framework. SMEs should assess internal capabilities through thorough evaluations of existing resources and competencies to identify gaps. They should engage with external networks by establishing strategic partnerships with academia, industry consortia, and government bodies. Integration of digital tools is vital, leveraging digital platforms to facilitate collaboration and streamline innovation processes. Focussing on talent development by investing in upskilling employees and fostering a culture of innovation is critical. Finally, monitoring and evaluation mechanisms should be implemented to regularly assess the effectiveness of open innovation practices and make necessary adjustments. By exploring the intricate dynamics of open innovation in SMEs, this study aims to offer valuable insights that inform both theoretical understanding and practical application, thereby contributing to the broader discourse on innovation management in resource-constrained environments.

Problem statement

Employees' negative behaviours impact the performance of individuals and have an unfavourable effect on organizational performance, thereby hindering the achievement of organizational goals (Barrett & Tsekouras, 2022). These detrimental behaviours fail to contribute to organizational performance improvements and the overall enhancement of SMEs' productivity and effectiveness. The departure of employees due to resignations or other causes has been widely acknowledged as a substantial hurdle for enterprises. Work alienation can lead to various adverse consequences including the disruption of interpersonal connections, diminished engagement in workplace endeavours, weakened trust within the organization, erosion of social bonds and a decline in employee motivation toward their tasks (Guo et al., 2023). When SMEs are unable to cultivate their own internal resources and skills, they may benefit from leveraging external ties for the resources they need (Alghababsheh, 2023). The resource accumulation theory asserts that a company's resources and capabilities are produced and managed internally whereas the network view posits that resources and capabilities can be accessed through external ties (Yao et al., 2024). The culture of open innovation in SMEs is still in its infancy as they often lack the knowledge, skills and abilities necessary to effectively conceive and implement fresh ideas (Bashir et al., 2023a). The engagement of SMEs in developing an innovation culture is noticeably lacking. SMEs need to enhance their capacity for open innovation to improve their business performance as integrative culture is regarded as a crucial component that might kick-start innovation (Isensee et al., 2020). Involvement in activities, adherence to standards, openness to change, and a sense of purpose are the four components that make up an integrative culture. This culture involves a strong sense of psychological ownership and engagement with organizations. Integrative culture is associated with adapting external ideas to ensure strategic direction and intention, goals, objectives and vision are achieved (Silva & Borsato, 2017). SMEs exhibit heightened sensitivity to unit transaction costs especially in domains where risk diversification is unfeasible, and they essentially assume the prevailing market prices. Within trade facilitation initiatives, it is essential to grasp that SMEs' fundamental preoccupations revolve around cost management. Challenges pertaining to transactional costs are amplified by variables such as heightened competitive pressures, evolving market demands, technological advancements and the limitations inherent to SMEs in terms of expertise, ingenuity, and innovative capabilities (Stevenson et al., 2023). The crisis involving questions of ethics in providing technology has led to hard lessons about the chasm between the actual processes of ethical behaviour in developing and selling technology to small companies. Commercial knowledge and sensitivities help companies boost their worth and differentiate themselves from the competition. However, the dangers of theft and leaks increase along with the intensity of market rivalry (Vitolla et al., 2019).

Objectives

This research aims to explore the correlation between Organizational Citizenship Behaviour (OCB) and open innovation (OI), specifically focussing on how OCB impacts open innovation across different appropriability regimes. It seeks to assess the influence of integrative culture on open innovation practices and its modulation by appropriability regimes. In addition, the study aims to analyze how the appropriability regime affects the connection between managerial ties and open innovation. Furthermore, it investigates the influence of transaction costs on the implementation of open innovation strategies and how appropriability regimes moderate the transaction costs-open innovation relationship. The study intends to comprehensively examine the joint effects of OCB, integrative culture, managerial relationships, and transaction costs on open innovation outcomes, considering the moderating effect of appropriability regime. By including appropriability regime, the research aims to highlight distinct approaches to innovation management. The goal is to identify optimal combinations of independent variables (OCB, integrative culture, managerial ties, and transaction costs) for specific appropriability regime, offering recommendations to enhance open innovation efforts for SMEs. The research seeks to provide empirical evidence on the relationships among the specified independent variables, the moderating factor, and open innovation. Through this investigation, the study aims to advance the understanding of factors influencing successful open innovation. By exploring the interactions between organizational behaviour, cultural factors, network relations, transaction costs, appropriability regimes, and open innovation, the research aims to provide a comprehensive view of the innovation process from the perspective of SMEs.

Literature review

Open innovation (OI)

Chesbrough and Bogers, (2014) endorsed the open innovation paradigm, emphasizing the use of internal and external knowledge flows to enhance innovation and the importance of collaboration, co-creation, and knowledge sharing across organizational boundaries. However, SMEs face unique challenges in managing OI processes due to limited budgets to enhance the internal innovation activities. SMEs do navigate OI by building close relationships with ecosystem participants, addressing challenges arising from mismatches between their business models and the OI paradigm (Bashir et al., 2023a). OI has evolved over time, encouraging researchers and practitioners to explore the concept from diverse perspectives. However, SMEs in manufacturing often lack enthusiasm for OI due to the rapid pace of technological advancements and the disruptive nature of digital innovation, which many perceive as overly complex and mature (Isensee et al., 2020). OI involves participation in innovation activities across all stages, not limited to research and development (R&D). It creates "porous" organizational boundaries, enabling resource sharing and collaboration between firms and external partners. This differs from the "closed" innovation model, where businesses rely solely on internal resources (Silva & Borsato, 2017). Through OI, firms can leverage external expertise, adopt advanced technologies, and license their innovations to others. The advantages of OI include accelerated product-market launches, access to cutting-edge technologies and acquisition of specialized expertise (Stevenson et al., 2023). Despite these benefits, SMEs must recognize that successful adoption of OI often requires time and strategic alignment. OI provides opportunities to strengthen innovation capacity and address commercial challenges. In addition, SMEs adopting OI can attract foreign investment, thereby enhancing capital inflows and contributing to national economic growth. Leadership, absorptive capacity, and organizational culture play pivotal roles in fostering OI, as highlighted in several studies. (Le & Lei, 2019), emphasize how organizational dynamics such as managerial ties and an integrative culture support OI adoption. Similarly, (Torkkeli et al., 2015) provide valuable insights into cultural enablers and barriers to innovation, stressing the significance of aligning organizational practices with OI goals.

While OI offers significant potential, SMEs face several barriers in adopting these practices. The study explores barriers to collaboration, investigating the challenges SMEs face in building and sustaining collaborative networks, particularly in managing the diverse interests of actors within their innovation ecosystems. Actor Network Theory (ANT) provides a framework to analyze these interdependencies (Choi et al., 2018). Another critical area is mitigating risks in partnerships, where Social Exchange Theory (SET) can guide studies on how SMEs manage risks related to intellectual property sharing, technology transfer and resource allocation in ensuring equitable exchanges (Kassa & Tsigu, 2022). Cultural and environmental factors also play a role and exploring how cultural disparities and geographic contexts influence SMEs' perceptions and capacities to engage in open innovation is essential. Both ANT and SET can elucidate the role of cultural and social dynamics in shaping collaboration outcomes. Additionally, understanding how SMEs sustain long-term OI practices to foster continuous innovation is crucial including examining how networks evolve and adapt over time.

Organizational Citizenship Behaviour (OCB)

Organizational Citizenship Behaviours (OCBs) refer to voluntary, discretionary actions by employees that exceed formal job requirements and contribute to the overall effectiveness of an organization and its members (Bogler & Somech, 2023; Organ, 1988). OCBs consist of behaviours driven by personal initiative and a genuine desire to assist others. Even these acts were not formally rewarded but they significantly enhances organizational culture and performance. Research has established a positive relationship between OCBs and outcomes such as improved organizational and group performance, lower turnover, reduced unproductive work habits, and increased productivity (Haass et al., 2023). Discretionary behaviour, a cornerstone of OCBs, refers to actions that are not explicitly recognized within an organization's formal reward system yet collectively contribute to its success (Elshaer et al., 2022; Organ et al., 2006). Five key dimensions of OCBs are commonly identified in the literature which is Altruism (helpful conduct), Conscientiousness (organizational compliance), Civic Virtue, Courtesy and Sportsmanship (Jonasson & Ingason, 2022). These dimensions encapsulate the essence of OCBs and their relevance to organizational performance, particularly in SMEs. Understanding and fostering OCBs within SMEs is essential as OCBs promote collaboration, employee engagement, and knowledge sharing, creating an environment conducive to open innovation. Employees who exhibit high levels of OCBs positively influence innovation performance as well as cultivating a corporate culture of adaptability and creativity (Grözinger et al., 2022). Strategically integrating OCBs within SMEs can strengthen open innovation initiatives by offering valuable insights for managerial decision making and policy development. Conversely, fostering an open innovation culture can stimulate OCBs by building a reciprocal relationship between organizational behaviour and innovation management.

Altruism

Altruism refers to the selfless desire to assist others without expecting anything in return (Knez et al., 2019). In the workplace, altruistic behaviours manifest as voluntary support to colleagues such as helping with challenging tasks or alleviating others' workloads (Rhoads & Marsh, 2023). Although altruism does not guarantee immediate benefits for the organization, its cumulative effects enhance productivity and harmony among employees (Barghouti et al., 2022). These altruism behaviours include assisting co-workers with tasks, mentoring, contributing extra effort to team projects etc. Altruistic behaviours foster trust, camaraderie, and collaboration which improve workplace morale and efficiency. A culture of altruism enhances job satisfaction and mental well-being, thus encouraging the exchange of ideas and fostering adaptability (Neaman et al., 2023). Moreover, organizations embracing altruism may adopt flatter hierarchies and collaborative practices, fostering innovation and continuous improvement (Mouser, 2023). Over time, these behaviours contribute to a more harmonious, productive and socially responsible workplace.

Conscientiousness

Conscientiousness is characterized by diligent, disciplined, and responsible behaviours that exceed formal job expectations (Ngqeza & Dhanpat, 2022). Employees demonstrating conscientiousness often exhibit punctuality, thoroughness and proactive attitude towards preventing workplace challenges. This trait is consistently associated with higher work performance that includes job proficiency, training efficiency and team productivity (Emich et al., 2023; Thu Hanh, 2022). Conscientious employees contribute to workplace harmony by inspiring professionalism and motivating their colleagues. Their meticulous approach enhances teamwork and reduces conflict, thereby positively influencing organizational culture and customer satisfaction (Nong et al., 2023; Tan et al., 2023). Furthermore, their adherence to standards and commitment to excellence foster a climate of accountability, continuous improvement and promoting long-term organizational success.

Civic virtue

Civic virtue involves proactive behaviours that reflect an employee's commitment to the organization's well-being and success (Graham et al., 1986; Organ, 1988). This includes staying informed about organizational matters, attending meetings, and representing the organization positively in both internal and external settings (Mishra et al., 2022). Employees exhibiting civic virtue demonstrate loyalty and a willingness to contribute beyond their prescribed roles, thereby fostering a sense of community within the workplace. Civic virtue enhances organizational performance by encouraging active participation and informed decision making (Oamen, 2023). Employees who demonstrate civic virtue often develop valuable skills such as effective communication and information processing, which benefits both their careers and the organization. These behaviours also foster a culture of responsible engagement as well as strengthening organizational resilience and adaptability.

Courtesy

Courtesy refers to proactive behaviours aimed at preventing workplace conflicts and fostering harmonious relationships (Law & Turner, 2008). This includes considerate actions such as informing colleagues about schedule changes, checking on their well-being or offering assistance with tasks (Kang & Hwang, 2023). Courtesy mitigates unnecessary stress and misunderstandings and promoting a respectful and inclusive workplace culture. A culture of courtesy enhances employee morale and engagement, fostering collaboration, trust and effective communication (Kuknor et al., 2023). By minimizing conflict and improving interpersonal dynamics, courtesy contributes to organizational productivity and employee well-being. Additionally, it encourages positive stakeholder relationships and ethical workplace practices as well as creating a supportive environment conducive to success.

Sportsmanship

Sportsmanship involves maintaining a positive attitude despite challenges or less-than-ideal working conditions (Masrohatin & Tobing, 2019). Employees exhibiting sportsmanship avoid unnecessary complaints and adapt constructively to changes or setbacks, thereby reducing the managerial burden and fostering a supportive workplace atmosphere (Fakhri et al., 2021). Sportsmanlike behaviour enhances workplace morale, teamwork and productivity. By promoting resilience and positivity, it contributes to a collaborative culture where employees can thrive even under pressure. Organizations that value sportsmanship benefit from reduced conflict, increased employee satisfaction and improved overall performances.

ANT and SET provide complementary frameworks for understanding OCBs. ANT highlights the interconnected networks of human and non-human actors that shape organizational behaviours, offering insights into how OCBs emerge and sustain within an organization's socio-technical ecosystem (Organ, 1988). SET, on the other hand underscores the reciprocal nature of interpersonal and organizational relationships, positing that OCBs arise from employees' perceptions of fairness, trust and mutual benefit (Organ et al., 2006). Therefore, SMEs should analyze the socio-technical networks that influence employee behaviours, including both human and non-human actors. In addition, SMEs must emphasize fairness and trust-building within organizational relationships. Providing equitable recognition, creating opportunities for professional growth, and maintaining transparent communication are critical to nurturing OCBs rooted in social exchange dynamics. By combining these theoretical perspectives, organizations can create an environment where OCBs flourish, leading to enhanced collaboration, innovation, and overall performance. Therefore, OCBs play a pivotal role in driving open innovation as discretionary employee behaviours such as sharing knowledge and assisting others beyond formal job roles create a collaborative environment conducive to innovation. As such OCB's role is critical in promoting a culture of innovation within SMEs, particularly in dynamic and resource-constrained environments where such behaviours can amplify organizational adaptability and creativity. (Lichtenthaler & Lichtenthaler, 2009) emphasized the importance of knowledge infrastructure and absorptive capacity in enabling SMEs to engage in open innovation. In addition, (Jasimuddin & Naqshbandi, 2019) acknowledged that infrastructure and absorptive capacity significantly influence open innovation in SMEs. A robust knowledge infrastructure enhances absorptive capacity, enabling SMEs to effectively leverage external knowledge for innovation. This places SMEs to leverage open innovation for improved performance and competitiveness. By fostering OCBs, SMEs can better leverage these capabilities, ensuring that employees actively contribute to knowledge exchanges and adapt external ideas to fit organizational needs.

H1: Organizational Citizenship Behaviour positively predicts the performance of SMEs with the adoption of open innovation.

Integrative culture (IC)

Integrative culture fosters collaboration and communication, breaking down silos within organizations to promote the assimilation of external ideas and knowledge into innovation processes (Azeem et al., 2021). Naqshbandi & Kamel, (2017) emphasized the importance of inclusiveness, adaptability and learning in creating a supportive environment for innovation as the study highlighted that an integrative culture not only encourages openness but also establishes a framework for organizations to absorb and apply external knowledge effectively. (Naqshbandi et al., 2015) examined the impact of organizational culture on open innovation, thus identifying that collaborative, knowledge sharing and flexible cultures are enablers of OI, whereas hierarchical and control-oriented cultures hinder adoption. In addition, (Naqshbandi & Kamel, 2017) had suggested that firms with a strong organizational culture that encourages knowledge sharing and learning are better able to absorb external knowledge and integrate it into innovation processes. The above study explored the relationship between organizational culture and open innovation, focussing on evidence from an emerging country. (Naqshbandi & Tabche, 2018) highlighted that absorptive capacity is essential for recognizing and applying external knowledge, while learning culture promotes continuous learning and knowledge sharing that enables successful OI. Various studies underscore the importance of fostering adaptive cultures to effectively leverage external knowledge and drive innovation. Drawing on ANT, integrative culture can be seen as a dynamic network where human and non-human actors including employees, technologies and organizational systems interact to co-create innovation (Aka, 2019). ANT explains how these interactions establish a cohesive and adaptive organizational structure that facilitates open innovation. At the same time, Social Exchange Theory (SET) emphasizes the role of trust, shared norms, and reciprocal relationships in promoting collaboration and knowledge-sharing within this culture (Kassa & Tsigu, 2022). Employees able to engage with innovation initiatives when they perceive mutual benefits and support from management (Torkkeli et al., 2009). ANT and SET provide a comprehensive theoretical framework for understanding how integrative culture drives OI adoption and enhances SME performance.

Employee development

Employee development programs align employee roles with organizational goals, enhance skills and foster commitment as these initiatives build trust in management and encourage reciprocal effort and loyalty from employees (Boccoli et al., 2022). Effective management addressing emotional needs strengthens this commitment and often creating a sense of moral obligation to stay especially when leaving is perceived as more costly than remaining. Employee development is a critical enabler of organizational growth and innovation (Kozhakhmet et al., 2022). Using Actor Network Theory, employee development programs can be viewed as networks of interconnected actors such as training systems, learning technologies, mentors and employees that collaboratively enhance workforce capabilities (Vickers et al., 2018). These interactions contribute to creating an environment that supports continuous learning and adaptability, essential for open innovation. Social Exchange Theory complements this perspective by explaining how development programs establish a sense of loyalty and mutual commitment between employees and the organization (Cropanzano et al., 2017). The interplay between these theoretical perspectives highlights how well-designed employee development programs can nurture a workforce that aligns with the goals of an integrative culture and open innovation.

Harmony

Organizational harmony involves aligning relationships among employees, between employees and management, and with external stakeholders (Tsui et al., 2006). Achieving this requires addressing workforce needs, fostering motivation, collaboration and a positive work environment. Clear objectives, defined responsibilities and structured processes are vital for cultivating harmony as harmony enhances workplace culture and facilitates collaboration, skill-sharing, and alignment with organizational goals, ultimately improving performance and effectiveness (Susila et al., 2023). Organizational harmony is vital for fostering a cohesive work environment that supports innovation and adaptability. Through the lens of Actor Network Theory, harmony can be seen as the outcome of balanced and functional interactions among various actors, including employees, management, technologies and external stakeholders (Soga et al., 2021). These interactions create a network where collaboration and knowledge-sharing are seamless, contributing to a harmonious organizational culture. Social Exchange Theory further enriches this understanding by highlighting the role of trust, fairness and reciprocity in achieving harmony (Zoller & Muldoon, 2019). The integration of these theories underscores how fostering harmony within an organization builds the foundation for a culture that embraces open innovation and drives performance.

Customer orientation

Customer orientation emphasizes prioritizing customer satisfaction by enhancing product quality and addressing challenges in delivering superior products as it fosters an integrative culture that promotes behaviours essential for providing exceptional value to consumers (Ismail, 2022). This approach focuses on building lasting customer relationships and demonstrates a commitment to organizational cultural transformation. Through Actor Network Theory, customer orientation can be conceptualized as a network of interactions between the organization and its employees, technologies and customers (Ozuem et al., 2021). These interactions enable the organization to gather process and act on customer insights thereby enhancing product quality and service delivery. Social Exchange Theory adds depth to this by explaining how reciprocal relationships with customers built on trust and mutual value exchange strengthen the organization's commitment to customer satisfaction (Abdou et al., 2022). These theoretical perspectives provide a robust framework for understanding how customer orientation contributes to an integrative culture that supports open innovation.

Innovative culture

An innovative culture promotes creativity, risk taking, and seamless integration of new ideas, supported by organizational encouragement and robust frameworks (Annamalah et al. 2018, Tsui et al. 2006). Successful open innovation adoption requires a unified innovative culture that fosters commitment, excellence and corporate citizenship that enables organizations to embrace and sustain innovation effectively (Alqudah et al., 2022). Innovative culture is fundamental to creating an environment where creativity, experimentation, and risk taking are encouraged. Actor Network Theory views innovative culture as a network of interdependent actors including employees, technological tools and organizational processes that collectively drive innovation (Aka, 2019). This network perspective highlights how interactions among these actors generate ideas, test solutions and implement innovations. Meanwhile, Social Exchange Theory explains how trust and mutual support within this culture encourage employees to propose and champion innovative ideas without fear of failure (Luqman et al., 2023). Together, ANT and SET provide a comprehensive understanding of how innovative culture supports the adoption of open innovation enhancing organizational performance.

H2: Integrative culture will positively predict the performance of SMEs with the adoption of open innovation

Managerial ties

Managerial ties, encompassing the connections of managers both within their organizations and with external political and governmental spheres are crucial for accessing external information and resources (Dai et al., 2022). Managerial relationships have been widely studied in terms of company performance, but their role in product innovation is less explored as these ties, both within business and political spheres, are crucial for accessing resources and information (Uzkurt et al., 2024). However, they involve both costs and benefits such as commitments, trust and support. The balance between investment and return in these relationships depends on power imbalances and trust. Overall, managerial ties shape institutional norms such as reciprocity and collaboration, and managers must adhere to these norms to navigate their networks effectively. While the strategic management literature extensively explores how managerial ties enhance firm performance, their role in product innovation warrants further investigation (Bashir et al., 2023b). The dynamics of managerial ties can be understood through Actor Network Theory and Social Exchange Theory. ANT elucidates how managerial ties form dynamic networks of interconnected actors' managers, organizations and institutions that facilitate resource sharing and innovation (Henttu-Aho et al., 2023). SET highlights how trust, reciprocity and mutual commitments influence the sustainability and benefits of these relationships (Huo et al., 2023).

Ties with other firm managers

Managerial ties with other firm managers are indispensable for developing market efficacy, seizing opportunities, and improving organizational performance as these relationships enable access to industry trends, market intelligence and best practices that bolster innovation and competitiveness (Ozanne et al., 2022). Building relationships with suppliers, customers, and competitors helps businesses seize opportunities, increase sales, and improve performance. Networking with external managers provides access to valuable resources and potential collaborations, driving efficiency, creativity and market expansion and this is very pertinent during economic instability whereby these networks can help businesses survive challenges (Clausen & Molden, 2024). Organizations should prioritize maintaining these strategic ties to enhance competitiveness and through collaborative efforts, businesses can achieve joint ventures by improving efficiency, enhancing creativity and expanding their customer base (Chung & Ho, 2024). Incorporating ANT, these ties represent a network of actors who collaborate to exchange knowledge, resources and opportunities, thereby creating a dynamic ecosystem that drives innovation. SET complements this by emphasizing the role of trust and reciprocal obligations in sustaining productive collaborations. By leveraging both theories, it becomes evident that managerial ties with other firm managers directly influence SMEs' strategic agility and long-term success (Möller & Halinen, 2000).

Ties with research institutions, universities, and business advisories

Collaboration with academic institutions, research organizations, and business advisories accelerates knowledge transfer and fosters innovation. Such partnerships provide access to ground-breaking research, advanced technologies, and cutting-edge methodologies, helping businesses create innovative products and processes (Chen, 2022). Advisory firms provide expertise in regulatory compliance and operational optimization, while educational institutions encourage a culture of continuous learning and skill development (Alghababsheh, 2023). These partnerships promote continuous learning, talent recruitment, and the sharing of industry insights. Collaboration with business advisers also improves efficiency, productivity, and strategic decision making by offering specialized knowledge as such ties also open doors to funding, market opportunities, and increased recognition, helping businesses expand their networks, broaden their client base, and enhance their market standing (Giachetti et al., 2023). From an ANT perspective, these collaborations form a network of interconnected actors' researchers, advisors and businesses who co-create value through shared knowledge and innovation. This network facilitates the assimilation and application of intellectual capital within SMEs. SET highlights the role of trust and mutual benefit in sustaining these partnerships. Integrating these theoretical lenses underscores the critical role of institutional ties in equipping SMEs to navigate industry challenges and achieve competitive advantages (Möller & Halinen, 2000).

Ties with officials (industry, government, and financial institutions)

Establishing strong ties with government officials, financial institutions and industry players is crucial for SMEs to gain institutional support, secure financial assistance and navigate regulatory challenges (Ouyang et al., 2023). These relationships enhance market access, provide resources and improve business survival and performance particularly in developing countries. Ties with government officials and financial institutions offer essential funding and stability, while connections with other organizations enable resource sharing and competitive advantages (Dong et al., 2022). For SMEs pursuing open innovation, these partnerships facilitate the acquisition and exploitation of knowledge, driving innovation as strong management ties improve the likelihood of attracting external resources, fostering new insights and breakthroughs (Sheldon & Kwon, 2023). ANT views these relationships as networks that enable SMEs to tap into institutional resources and expertise, fostering a collaborative environment conducive to innovation. SET further emphasizes the importance of trust and reciprocity in these interactions. By combining these theories, the strategic importance of managerial ties with institutional actors becomes apparent, particularly in driving SME performance through open innovation. (Naqshbandi, 2016) demonstrates that strong networks and absorptive capacity help firms innovate and gain a competitive advantage. However, (Naqshbandi & Kaur, 2014) caution that over-reliance on specific ties can lead to inefficiencies or dependency, as emphasized by SET's focus on balanced relationships. Therefore, when managerial ties are effectively managed, they serve as key drivers of open innovation by facilitating resource exchange while mitigating associated risks.

H3: Managerial ties will positively predict the performance of SMEs with the adoption of open innovation

Transactional costs

The field of Transaction Cost Economics (TCE) has greatly influenced organizational development and economic interaction. Neo-classical arguments from Williamson (1998) and additional development of transaction costs from Remneland-Wikhamn and Knights (2012) have established a dominant paradigm for understanding behaviour, markets, and organizations. According to Remneland-Wikhamn and Knights (2012), the existence of businesses stems from institutional transaction costs makes it more profitable to organize and control activities within an organizational hierarchy rather than relying solely on market mechanisms. While market linkages are often efficient, they incur additional costs such as information gathering, negotiation and monitoring. According to Actor Network Theory (ANT), organizational structures are not static, but they emerge from interactions between human and non-human actors such as institutional frameworks and technological tools. This perspective helps explain the dynamic interplay of actors that influences transaction costs. Social Exchange Theory (SET) highlights the reciprocal exchanges and trust-building mechanisms within partnerships that can either mitigate or amplify these costs. Notwithstanding the potential for higher transaction costs, industries and enterprises of all types are opening organizational barriers for collaboration to cope with external pressures (Nguyen & Nguyen, 2013). Through the lens of ANT, these costs are shaped by the complex networks of actors interacting across market and organizational boundaries. SET adds depth by emphasizing that trust and relationship building can offset opportunistic behaviour which is a key driver of transaction costs. For managers engaged in open and distributed innovation, coordination expenses and opportunistic behaviour are particularly significant challenges (Shao et al., 2022). Building on SET, strategic partnerships can help decrease these risks through reciprocal trust and shared objectives, although opportunism remains a concern for businesses seeking to maintain competitive advantages (Audretsch & Belitski, 2023).

Asset specificity

One of the most significant factors influencing production from a transaction cost perspective is asset specificity (Audretsch & Belitski, 2023). Assets may be human, such as specialized expertise, or physical, such as tailored machinery and equipment (Sarwar et al., 2023). Asset specificity significantly influences production and transaction costs, encompassing both physical assets, like machinery, and human resources with specialized expertise. Assets can be categorized as non-specific (standardized), idiosyncratic (highly tailored), or mixed mode (a combination of both) (Zheng et al., 2022). In open innovation, businesses often seek external resources to complement their own, necessitating careful consideration of the costs associated with transferring or sharing specific assets. Effective open innovation strategies must balance the benefits of external collaboration with the potential risks and transaction costs linked to asset specificity (Felsberger et al., 2022). Trust, risk, and the dynamics of such partnerships are shaped by the degree of asset specificity, making a thorough assessment of unique assets essential for managing open innovation successfully.

ANT's focus on the interconnectedness of human and non-human entities enriches the understanding of asset specificity as these specialized resources are embedded within networks that shape their value and functionality (Farida et al., 2024). SET emphasizes the mutual obligations and expectations that emerge when businesses share or transfer specific assets thereby underscoring the role of trust in these exchanges (Li et al., 2024). According to SET, the willingness to share assets depends on perceived reciprocal benefits and trust levels. However, ANT suggests that the degree of specificity influences how assets are integrated into broader networks, affecting trust and risk in partnerships. Strategies for open innovation must balance the benefits of collaboration with the potential drawbacks posed by asset-specific transaction costs (Felsberger et al., 2022). Businesses can successfully navigate these challenges by conducting rigorous assessments of the unique assets involved and leveraging ANT's insights into network dynamics to ensure alignment with their strategic goals.

Environmental uncertainty

Unstable technological advancements can disrupt production by increased transaction costs and businesses can mitigate by forecasting, organizing their workforce and leveraging OI to access diverse expertise and ideas (Chen et al., 2022). In uncertain contexts, OI helps organizations share risks, diversify knowledge sources, and accelerate innovation. By engaging with broader innovation ecosystems, businesses can better adapt to changing conditions, enhance scenario planning and maintain a competitive edge through innovative products and services as flexibility and external collaboration become vital in volatile environments (Atanassova & Bednar, 2022). Environmental uncertainty is shaped by the evolving relationships among technological, market, and regulatory actors from an ANT perspective. This interconnectedness means that businesses must continuously adjust their strategies to minimize transaction costs and remain competitive. SET reinforces this by emphasizing the importance of reciprocal exchanges in navigating uncertain environments, particularly through trust and mutual support within networks of collaborators. ANT's framework helps explain how businesses interact with this broader ecosystem, fostering adaptability and innovation whereas SET highlights the role of trust in these interactions, ensuring that partnerships remain productive despite volatile conditions (Stranieri et al., 2022). Organizations that effectively leverage external resources while maintaining strong internal networks can achieve competitive advantages by adapting to shifting market conditions and introducing innovative products and services (McManus, 2023).

Behavioural uncertainty

Behavioural uncertainty arises from the inability to predict the actions or intentions of external partners in transactions and strategic partnerships as it increases transaction costs when dealing with unreliable or opportunistic individuals (Falamarzi et al., 2023). Clear communication, shared goals, and open collaboration practices are key to fostering trust and mitigating behavioural ambiguity caused by cultural and decision-making differences Organizations must establish conflict resolution mechanisms and adaptable decision-making processes to address these challenges as managing behavioural uncertainty is critical for achieving open innovation goals and fostering productive, long-term partnerships. The concept of behavioural uncertainty highlights that transaction costs increase whenever interactions with unreliable or opportunistic parties occur (Huang et al., 2023). SET provides a framework for understanding how clear communication, common objectives, and conflict resolution methods can enhance trust and reduce disputes while ANT underscores the importance of aligning network dynamics to mitigate uncertainty. Businesses must adapt their decision-making processes to accommodate the needs and preferences of their external stakeholders, ensuring that open innovation initiatives remain productive and mutually beneficial (Amankwah-Amoah et al., 2022).

Technology competencies

Technological competencies enable businesses to recognize their strengths and enhance their effectiveness (Amankwah-Amoah et al., 2022). These competencies, which include credentials, experience, skills, and technological expertise, are crucial for engaging in open innovation. External technology adoption enables SMEs to acquire competencies through advanced tools and educational opportunities by providing returns on investment (Perifanis & Kitsios, 2023). Organization's technological skills help identify innovation gaps and opportunities for external collaboration, allowing businesses to seek partners with complementary capabilities (Razak et al., 2018). Companies with robust technological able to attract collaborators (Srisathan et al., 2023). SMEs with established competencies are better positioned to assess and mitigate risks in external partnerships. To excel in open innovation, SMEs must fully leverage their technological strengths to adapt and succeed in dynamic markets. ANT views technological competencies as active participants in innovation networks, shaping and being shaped by their interactions with other actors. SET complements this perspective by highlighting the reciprocal benefits of technological collaboration, such as access to cutting-edge tools and shared knowledge. Open innovation allows businesses to draw on a broader reservoir of scientific expertise and development tools. According to SET, partnerships that leverage technological competencies foster trust and long-term collaboration by demonstrating value through unique innovations, patents, and research outcomes (Perifanis & Kitsios, 2023). ANT's insights into network dynamics suggest that businesses must strategically position their technological competencies within the broader innovation ecosystem to maximize their potential. By aligning internal strengths with external opportunities, companies can navigate the complexities of open innovation and maintain competitive advantages.

Degree of competition

As competition intensifies, SMEs must form strategic partnerships to acquire the technological expertise necessary for sustaining competitive advantages (Latifi et al., 2022). Open innovation serves as a powerful approach, fostering collaboration in research and development networks to exchange information, develop new products, and drive breakthrough technologies. These collaborations provide a competitive edge by enabling cost-effective innovation and market disruption (Wu et al., 2023). Businesses in competitive markets can leverage open innovation to expand product lines or enter new markets while balancing self-reliance and collaboration (Yang & Hurmelinna-Laukkanen, 2022b). Thriving in such environments requires SMEs to strategically harness external resources and maintain adaptability. According to ANT, competition drives businesses to form alliances and integrate into innovation networks where diverse actors contribute to shared goals. SET emphasizes that reciprocal exchanges within these partnerships are vital for building trust and sustaining collaborative efforts. In highly competitive markets, open innovation provides an effective strategy for accessing new ideas, technologies and resources. ANT's focus on network dynamics explains how collaboration between diverse partners fosters the development of disruptive technologies that create competitive advantages. SET highlights the role of trust and mutual benefit in maintaining these partnerships, ensuring that businesses can innovate at low cost while broadening their market presence. By balancing self-preservation with collaborative efforts, businesses can thrive in competitive environments and maintain long-term success (Yang & Hurmelinna-Laukkanen, 2022b).

H4: Transactional cost will positively predict the performance of SMEs with the adoption of open innovation.

Appropriability regime

Appropriability refers to the ability of a resource owner to capture returns commensurate with its value as it also encompasses the capacity of various stakeholders to retain the financial benefits generated through the use of an invention. Strong appropriability regimes exist where the firm responsible for the development of an innovation is the primary beneficiary, whereas weak appropriability regimes are characterized by situations where the innovating firm benefits less than other stakeholders (Hurmelinna-Laukkanen & Puumalainen, 2007). An appropriability regime comprises the laws, policies, and strategies employed by organizations to safeguard inventions and intellectual property (IP). Mechanisms such as patents, copyrights, trade secrets and contracts significantly determine the extent to which organizations can control and profit from their innovations (Barchi & Greco, 2018). Appropriability regimes often emphasize ensuring that inventors maintain ownership over their inventions, enabling them to benefit financially and strategically from their creations. This approach remains highly relevant in today's industry, where protecting and benefiting from inventions is essential for fostering innovation (Hurmelinna-Laukkanen & Puumalainen, 2007). ANT provides a lens through which appropriability regimes can be examined by considering the dynamic networks of human and non-human actors involved in protecting and disseminating innovation. ANT explains how patents, policies, and technologies interact with inventors and organizations, shaping the appropriability regime. Similarly, SET emphasizes the interplay between economic incentives and social norms, highlighting how the balance of appropriability influences collaborative efforts in innovation ecosystems. Inventors continue to prioritize the security and ownership of their innovations and associated knowledge. Many are driven by anticipated private rewards, making the concept of appropriability central to motivating innovation. Risks associated with intellectual property leaks and theft are particularly significant in competitive markets, and therefore appropriability regimes influence innovation management by balancing the need for collaboration with the risks of losing proprietary information.

The impact of the appropriability regime on organizational citizenship behaviours (OCBs) and open innovation (OI)

The appropriability regime can significantly influence employees' willingness to engage in OCBs that promote innovation and intellectual property (IP) protection. Employees are more likely to report patentable ideas and participate in innovation efforts when they are assured that the organization's appropriability regime compensates and safeguards their contributions (Grimsby & Gulbrandsen, 2022). A strong appropriability regime reassures workers that their unique ideas will be protected and credited appropriately, reducing fears of IP theft and encouraging employees to share and innovate through OCBs (Bhatnagar et al., 2022). From the perspective of ANT, the appropriability regime serves as an essential actor within the organizational network. ANT highlights how appropriability mechanisms such as IP policies, patents, and collaboration agreements interact with employees, managers and external partners to shape innovation behaviours. SET offers insights into how appropriability regimes balance the economic incentives of protecting IP with the social norms that govern workplace collaboration. SET suggests that when employees perceive fairness in the distribution of rewards and recognition for their contributions, they are more likely to engage in behaviours that support open innovation. The interplay of economic rewards and social trust ensures that employees contribute meaningfully to innovation efforts without jeopardizing the organization's intellectual property. Proper management of IP rights is crucial for organizations embracing open innovation. Appropriability regimes can establish clear frameworks for protecting shared inventions in partnerships, ensuring that organizations benefit from external collaborations (Obaid, 2022). However, challenges arise when employees engaging in OCBs inadvertently divulge sensitive information to third parties, potentially harming the organization. A robust appropriability regime can mitigate these risks by offering protections that safeguard the organization's interests while encouraging employees to support open innovation projects (Hurmelinna-Laukkanen & Puumalainen, 2007). Appropriability regimes also influence how organizations select and interact with external partners. These regimes help set intellectual property-sharing agreements, ensuring that partnerships align with organizational goals while protecting key assets. As such, they foster an environment where innovative OCBs thrive as well as benefiting open innovation initiatives (Arsawan et al., 2022). Through the lens of ANT, the appropriability regime emerges as a critical component in the network of actors driving innovation, mediating relationships between employees, management and external collaborators. SET further explains how the appropriability regime aligns economic incentives with social norms, creating a supportive environment for OCBs and open innovation. Together, both theories provide a comprehensive understanding of how appropriability regimes moderate the relationship between OCBs and open innovation.

H5: Organizational citizenship behaviours and open innovation are moderated by the appropriability regime

The impact of the appropriability regime on integrative culture and open innovation

Collaboration, cooperation and a readiness to share knowledge and resources are hallmarks of an organization with an integrative culture (Wang et al., 2023). This culture emphasizes the removal of barriers between departments, promoting cross-departmental collaboration, and fostering cohesive teams with common objectives. In this context, the appropriability regime plays a pivotal role in shaping how the organization's culture supports or inhibits innovation. On one hand, a strict appropriability regime may prompt employees to safeguard their creative work for the company's benefit, driven by concerns over IP protection. This caution, however, can reduce the willingness to share knowledge, leading to a more siloed and less inclusive organizational culture, which may stifle innovation and inhibit the development of an integrative culture. According to ANT, the appropriability regime acts as an "actant" in the network, influencing the dynamics between human actors and non-human actors (Toroslu et al., 2023). When the appropriability regime emphasizes IP protection, it may create a more fragmented network of knowledge and innovation, limiting the flow of ideas and resources across the organization. Alternatively, an integrative culture can thrive if the appropriability regime includes measures to reward and safeguard employees' contributions to innovation while also promoting collaboration (Contigiani, 2023). SET can help explain this process, where employees view the exchange of ideas and knowledge as a benefit to both them and the organization. By providing the appropriate attribution and protection measures, organizations can motivate employees to freely contribute ideas, thereby fostering an open and integrative culture. Employees are more likely to engage in knowledge sharing because they perceive that their contributions will be appropriately rewarded and protected. SET also suggests that when the costs of sharing knowledge are perceived as low, employees are more willing to collaborate. However, a strict appropriability policy, focussed solely on protecting intellectual property, may discourage open innovation by creating barriers to knowledge exchange with external partners. Firms may be reluctant to engage in open collaboration if they fear losing control over proprietary knowledge. According to ANT, this reluctance to engage in broader networks due to stringent IP protection can isolate the organization from external actors that could provide valuable knowledge and innovation opportunities. On the other hand, a balanced appropriability regime that ensures adequate legal safeguards while allowing for the exchange of ideas can encourage open innovation. Ultimately, the appropriability regime is a critical component in managing innovation and organizational learning, as it helps shape the norms of cultural integration. As an organization's culture influences employees' attitudes and approaches to work, the appropriability regime serves as a supplementary element in reinforcing or impeding these cultural norms.

H6: Integrative culture and open innovation are moderated by the appropriability regime

The impact of the appropriability regime on managerial ties and open innovation

Executive boundary-spanning activities and the relationships with external entities that go along with them are described as managerial ties (Methot et al., 2022). Managerial ties are a type of social trade or capital. One definition of social capital is the goodwill generated by the network of social relationships and available for use in supporting collective action (Albis Salas et al., 2023). Effective open innovation practices are often shaped by both external networks and knowledge management strategies. (Naqshbandi et al., 2011) demonstrate that managerial ties play a pivotal role in accessing external knowledge, while appropriability regimes ensure the protection and effective utilization of this knowledge. These factors together enable organizations to balance openness and innovation security, enhancing their innovation outcomes. The ability of firms to protect the technology adopted from external parties depends on appropriability, even though the ties between these parties are cordial. ANT emphasizes the role of both human and non-human actors and in this case include firms, managers, technological resources, intellectual property and legal frameworks in shaping these relationships. In this context, the appropriability regime can be viewed as a key actant that structures the flow of knowledge, influencing how firms form and maintain managerial ties. When there is no possibility of appropriability, the seller will market the same concept to other parties, which deprives the company that bought the technology of the ability to engage in further innovation activities. The appropriation of the outcomes of inventive activity is crucial for SMEs because it allows them to benefit from the technologies they adopt externally (Nienaber et al., 2023). SET adds that these managerial ties are built on a cost–benefit exchange, where firms weigh the benefits of acquiring external resources. When businesses need to protect ideas, they may rely on external partners to help them succeed. In this scenario, managerial ties become critical because they facilitate the acquisition of external expertise, resources, and technologies. Communication benefits from stronger relationships between parties (Urresta-Vargas et al., 2023). According to SET, if the appropriability regime is conducive to protecting proprietary knowledge, firms will be more likely to form and maintain open innovation partnerships, sharing the costs and benefits of discoveries. This dynamic, influenced by both the social exchange and the technical components of the regime, leads to a higher likelihood of successful collaboration. How a business deals with management relationships and OI is profoundly affected by the appropriability regime (Liang et al., 2023), with firms adapting their innovation strategies and tie-building activities depending on the regime they operate under.

H7: Managerial ties and open innovation are moderated by the appropriability regime

The impact of the appropriability regime on transaction costs and open innovation

Transaction cost theory aims to explain when market and company actions are most likely to take place, but transaction costs have been disregarded as part of production costs (Remneland-Wikhamn & Knights, 2012). The transaction costs associated with open innovation may increase under a strict appropriability framework that provides extensive protections for intellectual property (Schäper et al., 2023). From ANT perspective, transaction costs are shaped by the configuration of actors involved in an innovation network. The appropriability regime as a key actor influences the complexity of interactions, the need for formal agreements, and the negotiation process all of which contribute to transaction costs. High transaction costs can arise when the protection of intellectual property is a significant concern, forcing firms to invest heavily in legal and contractual safeguards. An organization's ability to negotiate, minimize the need for elaborate contractual structures, and reduce monitoring and enforcement expenses may benefit from a greater willingness to share expertise and ideas with external partners. SET further explains this by suggesting that high transaction costs may discourage firms from engaging in open innovation, as the perceived costs of partnership formation may outweigh the expected benefits. OI may be stymied by these high transaction costs (Nora et al., 2022), which make firms hesitant to engage with external partners. If the costs of forming a partnership are too high, companies may be reluctant to seek external inspiration. OI can be encouraged when transaction costs are lowered. SET proposes that firms are more likely to collaborate when the perceived benefits, such as shared innovation and reduced costs that can outweigh the transaction costs. This can be achieved by streamlining collaborative procedures, increasing communication and developing standardized agreements that reduce the need for complex negotiations and enforceable contracts. The appropriability regime and its impact on transaction costs are intertwined, as a strict regime may raise transaction costs, thereby hindering successful open innovation. The tension between securing proprietary information and fostering external partnerships that drive innovation is a subject of ongoing study and practice for many researchers and professionals.

H8: Transactional cost and open innovation are moderated by the appropriability regime

Actor network theory (ANT)

ANT is a theoretical framework that emphasizes the influence of both human and non-human entities in shaping social processes. This approach suggests that even inanimate items, such as technology, can possess agency and play a role in the dynamics of social interaction and innovation. In ANT, an "actor" is defined as the "source of an action," which can be either a human or a non-human entity (Murdoch, 1997). The central idea of ANT is that actors do not operate in isolation but interact within networks where their actions are contingent upon the relationships and collaborations with other actors. These networks are shaped by social goals such as economic or professional objectives which drive the development of technology and the way it influences human interactions. Technology, in this sense, does not exert a one-way impact on people but rather affects how people interact with one another as it is embedded in the social fabric. ANT theorizes the formation and functioning of networks, exploring how connections emerge, how actors are enrolled into these networks and how networks may gain temporary stability or instability as new connections are made or removed. The presence or absence of any actor within the network can impact the overall network dynamics. Despite this, networks are fluid and constantly changing, as social realities are complex and flexible. This is why ANT focuses on studying how networks evolve, how they incorporate new actors and how they may reach a state of temporary equilibrium or become unstable as new relationships emerge (Marcon Nora et al., 2023; Ribes, 2018). ANT has been used to examine the role of key factors in socio-technological transformations across different industries, focussing on how technological and social elements come together to shape practices and outcomes in organizations and society (Engelsberger et al., 2023; Saleem & Raza, 2023). In business and innovation contexts, ANT helps explain how firms and their networks evolve over time and how external factors can drive changes within these networks. One important aspect of ANT is that while actors are often conditioned by their environments, they also have the potential to alter their surroundings. The adaptability is a key factor in the innovation process, where actors influence the innovation trajectory through their interactions and contributions within the network. Incorporating ANT into the understanding of OI provides a deeper insight into how innovation processes are influenced not only by human actors but also by the technologies and networks within which they operate. This perspective helps explain the complex and interconnected nature of innovation and how networks of actors collaborate to foster or inhibit technological advancement.

Social exchange theory (SET)

SET posits that individuals engage in social interactions based on a rational calculation of the costs and benefits involved. The theory suggests that people assess the potential rewards and sacrifices of their actions, and they are likely to engage in behaviours that they perceive as offering more benefits than costs. Conversely, if the perceived costs of a behaviour outweigh the benefits, individuals are less likely to participate in that behaviour. This cost–benefit analysis extends to various social interactions, including relationships and business partnerships (Barczak et al., 2022). According to SET, individuals evaluate relationships through a reciprocal process, where something of value is given and received. In social relationships, this reciprocity is crucial for sustaining interactions as individuals are more likely to continue engagements when they perceive a mutual exchange of benefits. The theory suggests that when individuals feel they are investing significant effort but receiving little in return, they may choose to disengage. This behaviour is fundamental in understanding how relationships evolve, particularly in contexts like collaboration, innovation and organizational settings. SET highlights how trust and reciprocity are essential for fostering long-term productive partnerships and collaborative efforts. In the context of open innovation, SET offers valuable insights into why individuals and organizations choose to collaborate. When partners in an innovation network perceive mutual benefits and trust the exchange process, they are more likely to engage in joint innovation activities. SET underscores the role of employee involvement climate in fostering collaboration. Empowering employees and creating an environment where contributions are valued help to establish an atmosphere of confidence and reciprocity, enhancing the likelihood of successful innovation outcomes (Catalano & Green, 2023). Strong employee involvement motivates participation and enhances open innovation effectiveness, with organizational culture playing a critical role in driving its success (Naqshbandi et al., 2019). By integrating SET, relational dynamics within organizations influence the willingness of individuals to engage in OI and as mutual benefits grow within these relationships, so too does the likelihood of successful co-creation, knowledge sharing, and innovative problem solving (Naqshbandi et al., 2019). Therefore SET, provides a strong theoretical foundation for understanding how relational dynamics shape collaborative innovation processes in organizational settings. (Naqshbandi et al., 2023) emphasized that a learning culture and knowledge sharing foster continuous learning, which enhances innovation and collaboration, while stressing the need to align HR strategies with open innovation for long-term success. The conceptual framework for the investigation is displayed in Fig. 1. The Fig. 1 framework shows how Organizational Citizenship Behaviours, Integrative Culture, Managerial Ties, and Transactional Costs influence Open Innovation, with Appropriability Regimes moderating these relationships.

Research methodology

A survey was undertaken among SMEs, and the research utilized a probability sampling approach with simple random sampling considered in accordance with quantitative methods in this study's design (White, 2022). Probability sampling approach is justified for surveys due to its ability to generate representative, unbiased, and generalizable

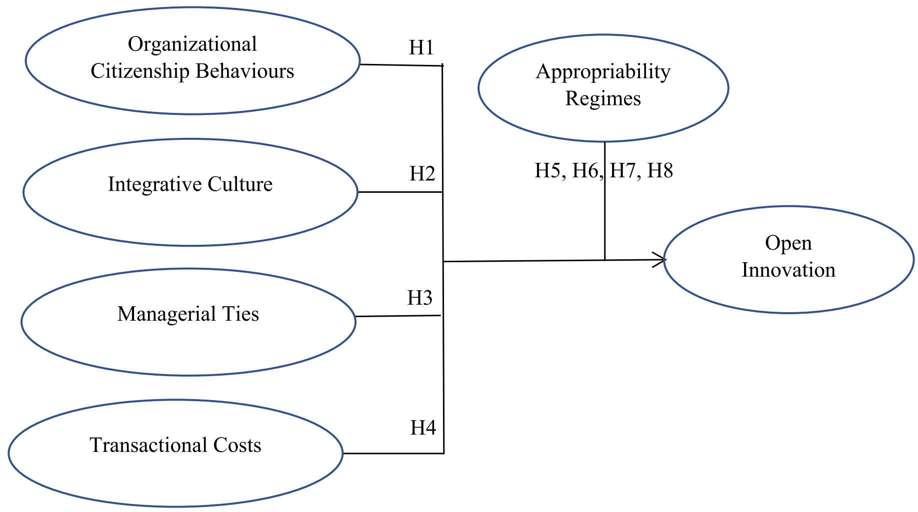

Fig. 1 Conceptual framework. Adapted from: Chesbrough and Bogers (2014), Naqshbandi et al. (2011), Remneland-Wikhamn and Knights (2012)

results, which are essential for drawing meaningful and valid conclusions about a known population. Probability sampling approaches like basic random sampling provide each population member an equal probability of selection. This creates a sample that closely resembles the population, making findings more representative and generalizable. Probability sampling reduces selection bias. Every person has an equal chance of being picked, reducing the likelihood of favouring certain groups or qualities and improving outcomes. Probability sampling underpins statistical inference. It helps researchers estimate population parameters based on sample characteristics, leading to more reliable results. Implementing simple random sampling is easy. Randomly picking people from the public does not need complicated techniques or understanding.

The systematic, standardized, and frequent way for gathering data from subjects who reflect the study's population is the survey method, which employs questionnaires to collect information from samples of individuals (Sarstedt et al., 2022). To ensure that a survey method is systematic and standardized in collecting data from subjects, establishing a clear procedures and protocols such as developing a well-structured survey instrument, conducting a pilot test of the survey, creating a comprehensive and accurate sampling frame and implementing randomization techniques in selecting survey participants and implementing quality control measures to monitor the data collection process as well as performing a thorough data cleaning to identify and correct errors. A thorough and systematic strategy, including many essential methods, was used to select a representative sample of SMEs for the research. A complete and up-to-date sample frame was established, listing all SMEs in the targeted population. This might entail getting official documents, industry association databases, or government databases on SMEs. These procedures guaranteed that every SME management had an equal chance of selection, reducing bias and boosting sample size. To ensure statistical significance, sample size was determined. The approach was carefully designed to assure feasibility and bias-freeness. Management and improvement of response rate were done. Data gathering was planned to assure authenticity. The cross-sectional design of this study necessitates selecting a representative sample of the population at a certain time and then using that data to answer the study's research questions. SME managers who own or operate in the industries were the primary respondents, utilizing a survey to gather primary data from SMEs. The sampling frame comprised SMEs registered in official government and industry databases, ensuring that the selected respondents were actively involved in management or strategic decision making. The surveys were personally delivered to the target respondents to ensure clarity regarding the questionnaire's intent and reduce the likelihood of incomplete responses.

Sampling frame

The sampling frame for this study was derived from government and industry databases, selected for their relevance to the targeted SMEs within the region. These databases were accessed through authorized channels, and verification was conducted to ensure the accuracy and recency of the listings. Criteria for inclusion required that SMEs were operational during the study period and relevant to the study's scope. Exclusion criteria included businesses that had ceased operations or were outside the defined industry sectors. The demographic distribution of the respondents, as outlined in Table 3, highlights their profiles in terms of age, gender, industry type, and company size. Efforts were made to ensure these demographics were representative of the larger SME population in the region. Although no quotas were imposed, the sample was evaluated post hoc to assess alignment with the broader population. Notably, statistical checks indicated a balanced representation in terms of gender, income, and race, reflective of the industry dynamics. Potential respondents were identified and selected through a randomized list generated from the sampling frame. Contact methods included email, phone calls, and, where applicable, in-person visits. Surveys were hand-delivered to ensure clarity in instructions and engagement. Locations for delivery were prioritized based on SME clustering in key industrial zones, ensuring broad geographic representation. To encourage participation, respondents were offered non-monetary incentives, such as a summary report of the study's findings. In addition, clear communication regarding the importance of their input ensured high levels of engagement and the accuracy of responses. The surveys were personally delivered by the research team, who also clarified questions to avoid misinterpretation. Each respondent was contacted up to three times, with follow-up visits to engage non-responders. Response integrity was maintained through cross-validation of survey answers with secondary data sources where available. The study adhered to strict ethical guidelines. Ethical approvals were obtained from the relevant institutional review board, and respondents were informed of the study's purpose, the voluntary nature of their participation, and the confidentiality of their data. Informed consent was secured in writing prior to participation. The data collection process was carried out systematically and standardized to ensure that the sample was representative of the population.

Confirmatory factor analysis (CFA) is a method employed to assess the validity of a hypothesized factor structure. On the other hand, partial least squares structural equation modelling (PLS-SEM) extends upon CFA by enabling the examination of both measurement models and relationships among latent variables. The Partial Least Squares Structural Equation Modelling (PLS-SEM) exhibits greater efficacy and adaptability but it's more complex to setup in terms of analysis compared to Confirmatory Factor Analysis (CFA). Therefore, confirmatory factor analysis was deployed to scrutinize the inherent factor arrangement within the questionnaire, accentuating the Likert scale attributes. Subsequently, Partial Least Squares—Structural Equation Modelling (PLS-SEM) was employed to assess the coherence and constancy of the collected data (Lakens, 2022).

The authors did not use any control variables in this study. The decision to exclude control variables was made based on the specific objectives of this research, which focussed on the direct relationships between the constructs. The exclusion of control variables in this study was a deliberate decision, supported by several theoretical, methodological, and contextual considerations. The primary focus of the research was on examining the direct relationships between the independent and dependent variables, and previous studies in similar contexts have shown minimal confounding effects from typical control variables, such as firm size, industry type, or geographic location. Additionally, the study's context SMEs operating in the manufacturing sector offered a relatively homogeneous sample, reducing the variability typically accounted for by control variables. Moreover, a review of relevant literature indicated that commonly used control variables had negligible effects on similar relationships in past research, reinforcing the decision to exclude them (Spector & Brannick, 2011). This approach also helped to simplify the analytical model, ensuring clarity and interpretability of the findings while maintaining focus on the primary variables of interest. Given the exploratory nature of the study, the primary objective was to establish initial insights rather than account for all potential influencing factors.

This study provides clear evidence of the validity and reliability of the measures used, as all constructs demonstrated high Cronbach's alpha values, exceeding the minimum threshold of 0.70. The methodology was robust, with carefully designed procedures for sampling, data collection, and analysis, ensuring that the findings are both valid and reliable. The 40 respondents in the pilot research, which represents an estimated 10% of the expected sample size for the full study which included managers and owners. Due to mistakes where respondents failed to reply to certain questions, only 386 out of the 500 organizations polled were useable. Table 1 demonstrates that all the study's variables had Cronbach's alpha values over the cut-off of 0.70, demonstrating the validity of the measures utilized in this investigation. Using a Smart-PLS technique, PLS-SEM was used to analyze the data. To clarify the specifics of the questionnaire, it was personally delivered to the managers, owners, and decision-making executives of the participating SMEs. Three hundred seventy-six (386) respondents were chosen for the sample size. In this study, 500 questionnaires were sent out, and only 386 valid responses or 77.2 percent

Table 1 Response from respondents

Response	Instruments
Questionnaires distributed	500
Questionnaires returned	398
Questionnaires excluded	22
Questionnaires useable	386
Response rate	77.2%

were chosen to analyze the data, as shown in Table 1. The data were examined using Smart-PLS 3 (SEM), which was based on the sample size (Sarstedt et al., 2022). The present study not much faces the challenges when using the Smart-PLS technique for data analysis. Only the model complexity and model specification were the challenging parts during the data analysis. However, these issues were resolved with theoretical foundation and carefully data preparation.

To ensure a seamless flow in the study approach, advice from academic experts and industry consultants was sought. Information obtained from managers who run or work for SMEs or from business leaders to research the actions of their workers, integrative culture, external ties, knowledge of transaction costs, and resistance to open innovation. Table 2 reliability test results demonstrate strong internal consistency across the variables measured in the study, with Cronbach's α values exceeding the acceptable threshold of 0.7. Organizational Citizenship Behaviour (OCB) with 14 items has a Cronbach's α of 0.813, indicating good reliability, while Interpersonal Communication (IC) with 10 items shows acceptable reliability at 0.746. Motivational Techniques (MT) and Organizational Innovation (OI) exhibit excellent reliability, with α values of 0.863 and 0.879 for 8 and 3 items, respectively. Team Collaboration (TC) has a Cronbach's α of 0.767 across 13 items, demonstrating consistent measurement, and Adaptability and Resilience (AR) shows good reliability at 0.821, even with only 2 items. The overall scale, encompassing all 50 items, achieves a strong Cronbach's α of 0.836, confirming the reliability of the constructs in the study.

Analysis

Table 3 presents the demographic characteristics of the respondents ($n = 386$). The majority are male (73.6%), with females accounting for 26.4%. In terms of race, most respondents are Chinese (72.3%), followed by Malay (13.5%), Indian (11.9%), and Others (2.3%). The largest age group falls between 31 and 40 years (31.1%), followed by 25–30 years (24.6%) and 41–50 years (22.0%), while smaller groups include 18–24 years (7.8%), 51–60 years (10.4%), and above 60 years (4.1%). Regarding education, 34.7% hold a degree, 23.1% completed high school, 14.5% are postgraduates, and 9.6% have primary education. For marital status, the majority are married (70.2%), while 27.7% are single, and 2.1% are divorced. On income, 33.4% earn between S$3000–S$6000, with others earning S$6001–S$9000 (36.8%), S$9001–S$12,000 (35.5%), or above S$12,001. Most respondents have working experience

Table 2 Reliability test

No	Variable	No. of items	Cronbach's α
1	OCB	14	0.813
2	IC	10	0.746
3	MT	8	0.863
4	TC	13	0.767
5	AR	2	0.821
6	OI	3	0.879
7	Total	50	0.836

Table 3 Demographic variables ($n = 386$)

Item	Description	Frequency	Percentage
Gender	Male	284	73.6
	Female	102	26.4
Race	Chinese	279	72.3
	Malay	52	13.5
	Indian	46	11.9
	Others	9	2.3
Age	18–24 years	30	7.8
	25–30 years	95	24.6
	31–40 years	120	31.1
	41–50 years	85	22.0
	51–60 years	10.4	10.4
	Above 60 years	16	4.1
Education	Primary	37	9.6
	High School	89	23.1
	Degree	134	34.7
	Postgraduate	56	14.5
Married	Single	107	27.7
	Married	271	70.2
	Divorced	8	2.1
Export	Yes	129	33.4
Income	No	257	66.6
	S$3000- S$6000	47	12.2
Working Experience	S$6001- S$9000	142	36.8
	S$9001- S$12,000	137	35.5
	Above S$12,001	60	15.5
	1–5 months/	22	8.8
	6 months – 1 year	23	9.2
	1–3 years	78	31.1
	4–5 years	60	23.9
	Over 5 years	68	27.1
Operation	1–5 years	53	13.7
	6–10 years	142	36.8
	Above 11 years	191	49.5
Turnover	Less than S$500,000	56	14.5
	S$500,001 S$1,000,000	171	44,3
	S$100,001 – S$10,000,000	113	29.3
	Above S$10,000,001	46	11.9

of 1–3 years (31.1%), followed by over 5 years (27.1%), while others have 4–5 years (23.9%), 6 months–1 year (9.2%), or 1–5 months (8.8%). In terms of company operation duration, 49.5% of companies have been in operation for over 11 years, 36.8% for 6–10 years, and 13.7% for 1–5 years. For company turnover, 44.3% report revenues of S$500,001–S$1,000,000, followed by S$1,000,001–S$10,000,000 (29.3%), less than S$500,000 (14.5%), and above S$10,000,001 (11.9%). This summary highlights key demographic and operational insights from the respondents.

Table 4 presents the descriptive statistics for the 386 respondents, indicating generally positive perceptions across variables. Organizational Citizenship Behaviour (OCB), Integrative Culture (IC), and Managerial Ties (MT) show high means of 4.026, 4.004, and 4.122, respectively, with moderate variability. Transaction Costs

Table 4 Mean and standard deviation (*N* = 386)

Variables	Mean	Std. Deviation
OCB	4.026	0.597
IC	4.004	0.576
MT	4.122	0.615
TC	3.520	0.528
AR	3.181	0.824
OI	3.957	0.261

Table 5 Multicollinearity Test

Constructs	Dependent Variable	Collinearity Statistics	
		Tolerance	VIF
MT	OI	0.837	1.054
IC		0.834	1.059
OCB		0.872	1.017
TC		0.867	1.023

(TC) has a lower mean of 3.520, while Appropriability Regime (AR) records the lowest mean of 3.181 and higher variability (SD = 0.824), suggesting more diverse responses. Open Innovation (OI) demonstrates a mean of 3.957 with the lowest variability (SD = 0.261), indicating strong agreement among participants. Overall, the results highlight positive yet varied perceptions.

Table 5 presents the results of a multicollinearity test to determine whether the independent variables (Managerial Ties [MT], Integrative Culture [IC], Organizational Citizenship Behaviour [OCB], and Transaction Costs [TC]) are highly correlated when predicting the dependent variable, Open Innovation (OI). The test uses Tolerance and Variance Inflation Factor (VIF) as indicators to assess multicollinearity. The Tolerance values for all constructs range from 0.834 to 0.872, which are close to 1. This indicates that the variability of each independent variable is largely independent of the others, suggesting low multicollinearity. Similarly, the VIF values for all constructs are between 1.017 and 1.059, far below the threshold of 10 that would indicate significant multicollinearity issues. These results confirm that there are no concerns about excessive multicollinearity among the independent variables.

Homoscedasticity

Table 6 presents the results of Levene's Test for Equality of Variances, used to assess the homogeneity of variances for the variables. The nonmetric variables were used to calculate Levene's test for the metric variables. According to Table 6, no variables were statistically significant (p value > 0.05), indicating that the assumption of homogeneity of variance, which states that the variance of a given variable remains constant across samples, was not violated.

Table 6 Homogeneity test

Variables	Levene Statistic	df1	df2	Sig
MT	12.679	1	384	0.194
IC	4.724	1	384	0.162
OCB	2.537	1	384	0.386
OI	1.969	1	384	0.621
AR	1.648	1	384	0.538

Table 7 Explanation of variance

Component	Initial Eigen values			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	12.72	20.20	20.20	12.72	20.20	20.20	12.51	19.90	19.90
2	7.40	11.74	32.01	7.40	11.74	32.01	7.21	11.50	31.33
3	7.02	11.15	43.10	7.00	11.15	43.10	5.50	8.70	40.00
4	5.02	8.00	51.05	5.00	8.00	51.05	5.00	7.90	47.90
5	3.22	5.11	56.20	3.22	5.11	56.20	4.91	7.80	55.70
6	2.42	3.83	60.00	2.42	3.83	60.00	2.70	4.30	60.00

Method: principal component analysis

Common method bias

Table 7 provides the results of the Explanation of Variance analysis using Principal Component Analysis (PCA). This analysis identifies the components that account for the variance in the dataset, focussing on the total variance explained by each component. The Initial Eigenvalues indicate the variance captured by each component before extraction. The first six components collectively account for 60% of the total variance, with the first component contributing the most (20.20%) and subsequent components explaining progressively less. The Extraction Sums of Squared Loadings show the variance captured by the components retained after extraction, which remains the same as the initial eigenvalues since the PCA method used retains components with eigenvalues greater than 1. The Rotation Sums of Squared Loadings represent the variance explained after the components are rotated for better interpretability. Rotation redistributes the variance among the components, slightly reducing the variance of the first few components while maintaining the cumulative variance at 60%. The first component explains 19.90%, and the sixth component explains 4.30% after rotation. The six components retained in the analysis collectively explain 60% of the total variance in the data, with the rotation improving the clarity of the component structure without altering the overall variance explained.

Assessment the measurement model

In evaluating the measurement model for this study, standardized factor loadings were examined to determine the reliability and significance of the observed variables. In this study, a stricter threshold of 0.7 was applied. Any items with standardized factor loadings below 0.7 were eliminated to refine the measurement model and improve overall model fit, as this threshold ensures that the retained indicators strongly represent the underlying latent constructs. This process ensures that the constructs meet the required levels of convergent validity and internal consistency. The initial model, before incorporating the moderating variable, was assessed to ensure all constructs demonstrated acceptable levels of reliability and validity. As shown in Fig. 2, the measurement model was first tested by examining factor loadings, composite reliability (CR), and average variance extracted (AVE) for each construct. The refined measurement model served as a robust foundation for further analysis, such as structural model testing and the incorporation of moderating variables.

Reliability of first-order indicators and internal consistency

All composite reliability (CR) values and factor loadings exceeded the threshold of 0.7, as shown in Tables 15–20. The Average Variance Extracted (AVE) values also met the acceptable criterion of greater than 0.5. Cronbach's alpha and composite reliability for all study constructs surpassed 0.7, affirming their reliability. Additionally, the AVE for each construct exceeded 0.5, indicating satisfactory convergent validity. A detailed summary of the reflective constructs, including AVE, Cronbach's alpha, CR, and confirmatory factor analysis, is provided in Tables 15–20 (refer to Appendix 1).

Table 8 displayed the measurement model for the AVE and CR second-order constructs. Table 8 demonstrated that all study constructs are valid because their CRs are all more than 0.70. Additionally, the AVE of every research construct is greater than 0.5, therefore this result denotes adequate convergent validity.

Table 9 illustrates the correlation matrix for first-order constructs with second-order constructs, providing insights into the relationships between these constructs. Correlation values range from − 1 to + 1, where values close to + 1 or − 1 indicate stronger

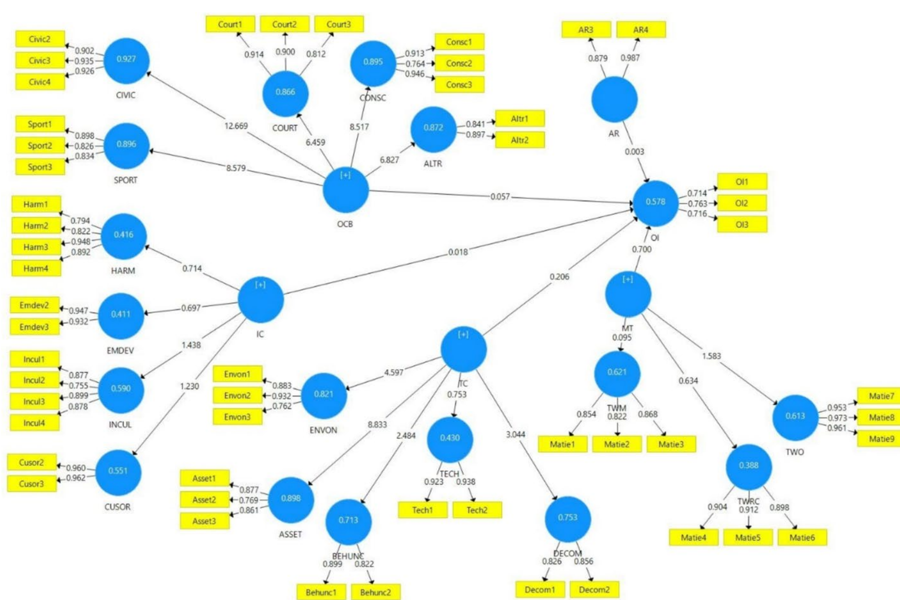

Fig. 2 Measurement model 1

Table 8 Measurement for 2nd-order constructs

Constructs	Variables	Composite Reliability (CR)	AVE
Organizational Citizenship Behaviours	Altruism	0.944	0.728
	Conscientiousness		
	Courtesy		
	Civic virtue		
	Sportsmanship		
Organization Culture	Employee Development	0.938	0.573
	Harmony		
	Customer Orientation		
	Innovative Culture		
Managerial Ties	Ties with Managers	0.885	0.513
	Ties with Research Centres		
	Ties with Officials		
Transactional costs	Asset Specificity	0.896	0.521
	Environmental Uncertainty		
	Behavioural Uncertainty		
	Technology Competency		
	Degree of Competition		

Table 9 Correlation of 1st-order constructs with 2nd-Order Constructs

	Constructs	1	2	3	4	5	6
1	AR	1.000					
2	MT	− 0.013	1.000				
3	IC	− 0.076	0.136	1.000			
4	OCB	− 0.467	0.026	0.078	1.000		
5	OI	− 0.173	0.621	0.198	0.296	1.000	
6	TC	0.068	0.027	− 0.21	− 0.095	0.319	1.000

relationships, and values near 0 suggest weak or no relationship. For first-order- and second-order variables in Table 9, diagonal elements are bigger than off-diagonal elements in the same row and column. A sufficient level of discriminant validity is shown by the square correlations for each construct being lower than the average variance retrieved by the indicators assessing that construct. Overall, the measurement model showed sufficient discriminant and convergent validity.

Discriminant validity

Table 10 shows that first-order- and second-order diagonal elements are bigger than adjacent off-diagonal components in the same row and column. A sufficient level of discriminant validity is shown by the square correlations for each construct being lower than the average variance retrieved by the indicators assessing that construct. Overall, the measurement model showed sufficient discriminant and convergent validity. According to Table 10, the bolded diagonal elements at the relevant rows and columns are larger than the off-diagonal components (square root of AVE). The Fornell-Larcker Criterion confirms that each construct satisfies the requirements for discriminant

Table 10 Discriminant Validity (Fornell–Larcker Method) for 2nd-Order Constructs with 1st-order Constructs

	Main construct	1	2	3	4	5	6
1	AR	**0.944**					
2	MT	− 0.014	**0.812**				
3	IC	− 0.072	0.143	**0.833**			
4	OCB	− 0.516	0.027	0.078	**0.846**		
5	OI	− 0.212	0.631	0.224	0.332	**0.785**	
6	TC	0.076	0.054	− 0.151	− 0.095	0.318	**0.747**

Bold diagonal values represent the square root of AVE, which are greater than the corresponding off-diagonal correlations, thus confirming discriminant validity based on the Fornell–Larcker Criterion

AR Appropriability regime; *MT* Managerial ties; *IC* Integrative culture; *OCB* Organizational citizenship behaviours; *OI* Open innovation; *TC* Transactional costs

Table 11 Discriminant validity Heterotrait–Monotrait Ratio (HTMT) Method for 2nd-order constructs with 1st-order constructs

	Main construct	1	2	3	4	5	6
1	AR						
2	MT	0.108					
3	IC	0.112	0.146				
4	OCB	0.750	0.052	0.115			
5	OI	0.319	0.300	0.089	0.383		
6	TC	0.179	0.228	0.091	0.212	0.418	

validity. The diagonal values (square root of AVE) being greater than the off-diagonal correlations ensure that the constructs are conceptually distinct, and no significant overlap exists between them in the measurement model.

Table 11 presents the Discriminant Validity results using the Heterotrait-Monotrait Ratio (HTMT) method for the second-order constructs with first-order constructs. Discriminant validity assesses the extent to which a construct is truly distinct from other constructs in the model. The HTMT values are compared to the commonly accepted threshold of 0.85 (or a stricter threshold of 0.90). Overall, the HTMT values for all constructs fall within the acceptable range, demonstrating that the model meets the discriminant validity criteria. This ensures that each construct measures a unique aspect of the theoretical framework.

Assessment of structural model

Table 12 indicates that the inclusion of the appropriability regime as a moderator increases the R-squared (R^2) value for open innovation from 0.578 in Model 1 (without the moderator) to 0.608 in Model 2 (with the moderator), indicating a substantial proportion of variance explained. The moderating factors exhibit a positive relationship with open innovation, consistent with the initial predictions.

Figure 3 illustrates Measuring Model 2, which presents the structural framework and indicators used to evaluate the key variables in the study. This model incorporates both latent constructs and their observable indicators to assess construct validity (how well

Table 12 Comparison between main effects and interaction effects (Model 1 & 2)

Path	Model 1 (without moderation)			Model 2 (with moderation)		
	β	R^2	Q^2	β	R^2	Q^2
MT—OI	0.095	0.578	0.227	0.174	0.608	0.296
IC—OI	0.018			0.271		
OCB—OI	0.057			0.135		
TC—OI	0.206			0.339		
MT × AR—OI	–			0.110		
IC × AR—OI	–			0.068		
OCB × AR—OI	–			0.087		
TC × AR—OI	–			0.151		

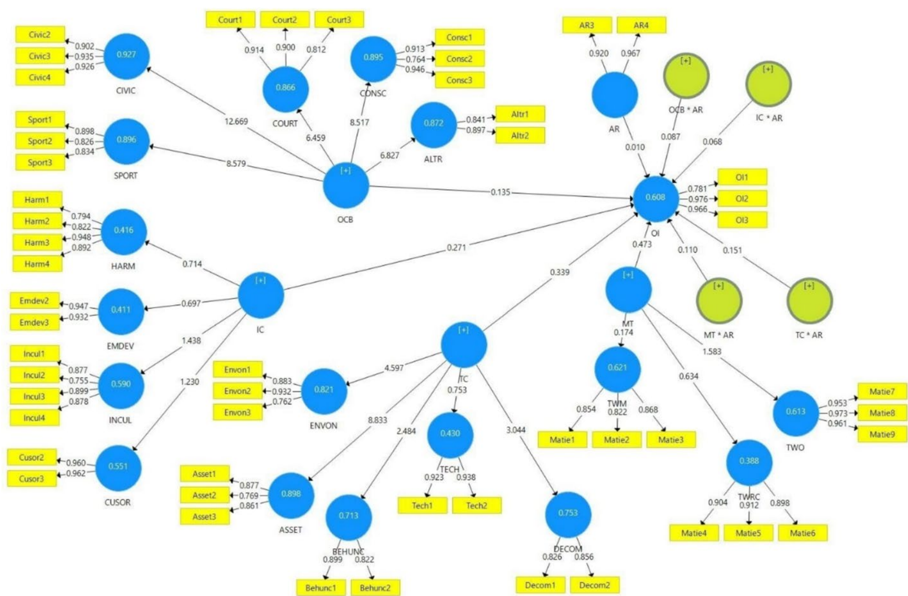

Fig. 3 Measuring model 2

the indicators represent the constructs) and reliability (consistency of the indicators). In this iteration, the moderating variable has been integrated into the model to examine its influence on the relationships between specific latent constructs.

Following the bootstrapping procedure, the relationships between managerial ties, transaction costs, and the appropriability regime were confirmed. Bootstrapping, a resampling technique commonly used in Partial Least Squares Structural Equation Modelling (PLS-SEM), evaluates the reliability and validity of model estimates. This method facilitates the assessment of model fit statistics, path coefficients, and standard errors. Additionally, bootstrapping plays a crucial role in addressing issues related to sample size, non-normality, and complex model structures, ensuring robust and reliable results (see Fig. 4).

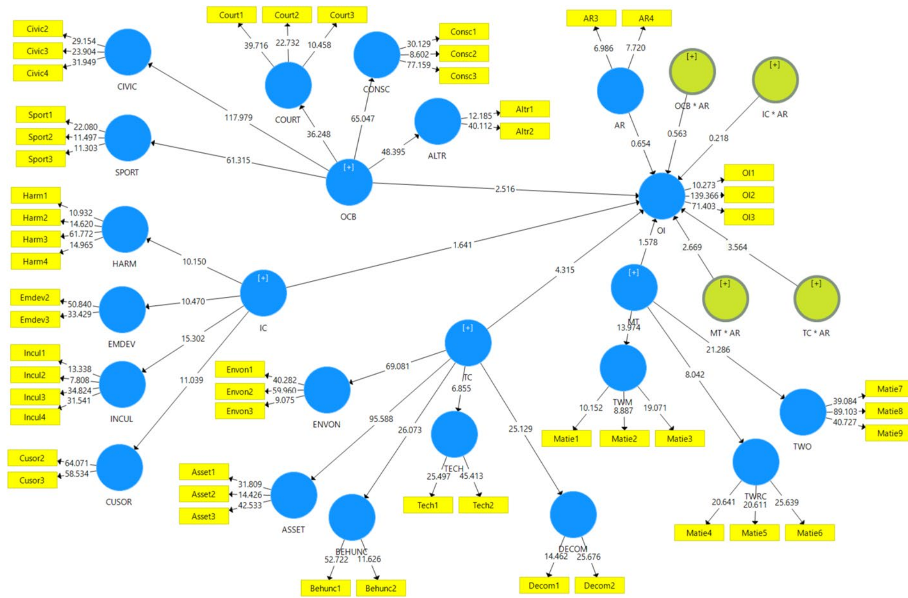

Fig. 4 Boot strapping result

Research hypotheses

The t-ratio must be more than 2 or the P value must be less than the significance threshold (0.01, 0.05, or 0.1). The coefficient of determination measures how much variance in the dependent variable the independent variable explains (s). 0.263 suggests that independent variables explain 26.3% of the dependent variable's variance. According to Table 13, the adoption of open innovation was positively correlated with organizational citizenship behaviour, integrative culture, management relationships, and transaction costs. Transactional expenses, however, have a significant influence on corporate processes (see Table 13).

Table 14 demonstrates that neither the appropriability regime nor the integrative Culture had any influence over the correlations between corporate citizenship behaviours

Table 13 Path coefficient test (model 1)

	Relationship	Coefficient	t-statistic	P value	Supported
H1	OCB—OI	0.263	2.526	0.012	Supported
H2	IC—OI	0.101	4.805	0.047	Supported
H3	MT—OI	0.664	6.130	0.042	Supported
H4	TC—OI	0.462	4.382	0.000	Supported

Table 14 Analysing the importance of the path coefficient of the moderating impact of the appropriability regime (model 2)

	Relationship (interaction effect)	Coefficient	t-statistic	P value	Supported
H5	OCB × AR—OI	0.062	0.562	0.603	Not Supported
H6	IC × AR—OI	0.089	0.223	0.823	Not Supported
H7	MT × AR—OI	0.199	2.199	0.030	Supported
H8	TC × AR—OI	0.078	2.854	0.042	Supported

and the adoption of open innovation. The links between management ties and the adoption of open innovation are, however, moderated by the appropriability regime. In Model 2, all path coefficients remain relevant, similar to Model 1. Additionally, Model 2 highlights the significance of the two interaction parameters TC x AR and MT x AR.

Discussion

This study highlights the positive effects of OCB on enhancing employees' creative and logical reasoning abilities, essential for fostering OI. While planning and goal setting are important, the emphasis is placed on human resources and creating a balance between integrative culture and knowledge management to improve SME practices. Managerial ties and OI are shown to positively influence SMEs' adoption of OI, offering insights into business, academia, and government collaboration to boost productivity. Transactional costs are identified as facilitating the shift from closed to OI by reducing manufacturing expenses, with technology playing a crucial role in overcoming these costs. The study also demonstrates that appropriability policy significantly impacts OI, further supporting the relationship between the two.

Theoretical implications

The development of Singapore as an industrialized nation has been closely tied to technological improvement and the adoption of innovative frameworks. This study offers a robust framework model for the execution of open innovation strategies, specifically targeting SMEs in Singapore and other emerging economies. Given the increasing interest of both academics and practitioners in open innovation, the proposed model integrates behavioural and costs factors that significantly influence OI adoption. The findings emphasize the critical role of OCB in fostering an environment conducive to OI. OCB encompasses the proactive engagement of employees, external partners and digital tools, contributing to the mobilization of internal and external actors. This study aligns with previous literature that highlights OCB's role in knowledge sharing, collaboration and co-creation of innovations (Organ et al. 2006; Podsakoff et al. 2000). OCB facilitates the development of social capital through trust and reciprocity, which are critical for successful knowledge flow and resource exchange within innovation networks (Nahapiet & Ghoshal, 1998). The results also show that OCB activities extend beyond internal collaboration to involve intermediaries, such as boundary spanners or facilitators, who translate knowledge and resources between different actors. This aligns with findings by Hargadon and Sutton (1997), who emphasize the role of knowledge brokers in innovation processes. Furthermore, reciprocal behaviours observed in OCB, where employees share knowledge with external partners expecting mutual benefits, were consistent with the principles of SET (Blau, 1964). By confirming the role of OCB in enhancing open innovation practices, this study expands on the existing literature by demonstrating how OCB contributes to the accumulation of social capital, which serves as the foundation for collaboration and resource mobilization. This is a notable extension of the work by Chesbrough (2006), who emphasized the importance of external partnerships in OI.

The findings highlight the importance of an integrative culture in bridging gaps between heterogeneous actors, facilitating communication and fostering a shared sense of purpose. Integrative culture creates a conducive environment for effective knowledge sharing, which aligns with the findings of Nonaka et al. (1996) on the role of organizational culture in knowledge creation. By mobilizing employees to participate in OI initiatives, integrative culture supports the development of collaborative platforms and idea management systems, which are essential for innovation processes (Tushman & O'Reilly, 2006). This study contributes to the understanding of integrative culture by emphasizing its role in promoting reciprocity and social exchange within organizations. The results align with prior research (Tsai & Ghoshal, 1998), which found that social capital facilitates the exchange of knowledge and resources in innovation networks. Furthermore, the integration of ANT provides a novel perspective on how non-human actants such as digital tools and platforms that interacts with human actors to support open innovation adoption (Latour, 2007). In contrast to studies that focus solely on competition as a driver of innovation, our findings reveal that cooperation within an integrative culture can be equally effective in fostering OI. This dual perspective of cooperation and competition contributes to a more nuanced understanding of open innovation strategies, addressing a gap highlighted in the literature (Gnyawali & Park, 2011).

Managerial ties, both internal and external play a significant role in bridging diverse actors and facilitating knowledge flow, resource mobilization and idea dissemination. This study reinforces previous findings by Acs et al. (2015) on the role of managerial networks in enhancing OI. By leveraging SET and ANT, our results demonstrate that managerial ties mobilize actors and resources, creating trust and reciprocal support within innovation networks as this finding also align with (How It Works, 1998). Transaction Cost Economics (TCE), which suggests that the choice between market-based transactions and hierarchical governance depends on transaction-specific factors. This study provides insights into how these decisions influence the coordination of external partners in OI initiatives. Specifically, it is found that adherence to norms within the network reduce transaction costs and enhance collaboration, supporting previous research by Dyer and Singh (1998). Furthermore, the findings highlight the role of appropriability regimes in influencing the links between innovativeness, managerial ties, and transaction costs. This expands on prior literature by demonstrating that strong appropriability regimes enable firms to balance the trade-off between knowledge sharing and protection, thereby fostering OI adoption (Teece, 1986). These insights contribute to the broader innovation literature by offering a multi-theoretical lens to understand the mechanisms through which SMEs able to adopt and implement OI practices. Future research can build on this framework to explore the role of digital tools, intermediaries, and governance structures in facilitating OI in other organizational and national contexts.

Practical implications

The findings support the growing body of literature emphasizing the importance of OI particularly in SMEs, to drive performance and profitability (Laursen and Salter, 2006; Vanhaverbeke, 2017). OCBs, defined as voluntary, extra-role behaviours that benefit the organization (Organ, 1988), play a crucial role in fostering a culture of OI. These behaviours, which include cooperation, helping others, and engaging in voluntary activities, can significantly enhance innovation processes by facilitating knowledge sharing and collaboration across organizational boundaries. As noted by Podsakoff et al. (2000), promoting OCBs can help SMEs reduce bottlenecks and improve resource utilization, thus creating a positive feedback loop that accelerates innovation. To leverage these behaviours effectively, SMEs must establish a supportive organizational culture that recognizes and rewards OCBs (Ali et al., 2023). By doing so, management can create an environment that nurtures collaboration and boosts innovation outcomes. The practical implication of these findings suggests that SMEs should integrate OCBs focussed initiatives into their strategic planning, providing training programs that emphasize the importance of voluntary behaviours for organizational success.

The importance of an integrative culture in facilitating OI is underscored by our findings, aligning with the work of Barjak and Heimsch (2023), who argue that organizational culture must actively support the adoption of external knowledge. An integrative culture prioritizes learning, adaptability, and openness to external ideas, all of which are essential for the successful implementation of OI. This type of culture can overcome the structural challenges SMEs often face when integrating external innovations and adopting new practices (Albats et al., 2023). Moreover, as suggested by Teece (1986) firms with integrative cultures can more effectively absorb external innovations, enabling them to remain competitive in dynamic markets. Our findings reinforce this by highlighting how an integrative culture empowers employees to make autonomous decisions and explore new ideas, which is essential for the success of OI. However, SMEs may face challenges in fostering such a culture due to resource constraints and organizational inertia. Therefore, it is recommended that SMEs focus on creating a culture that encourages continuous learning and provides employees with the autonomy to experiment and innovate.

The role of managerial ties in fostering OI is well documented in the literature, and our findings align with previous research on the importance of strategic partnerships (Clausen & Molden, 2024)). Strong managerial ties to external parties including suppliers, customers and research institutions, provide SMEs with access to valuable resources, industry insights and innovative technologies. Managers who actively cultivate these relationships can secure external resources that are critical for innovation. Furthermore, as (How It Works, 1998) notes in his theory of transaction costs, SMEs can reduce uncertainty and information asymmetry through strategic alliances and collaborations with external partners. This, in turn, allows them to improve decision making and gain competitive advantages. The practical implications suggest that SMEs should prioritize building and maintaining strong external ties, focussing on collaborations that enhance their innovation capabilities. Managerial leadership plays a key role in identifying and establishing these relationships, which can be critical to the successful adoption of OI practices.

Transaction cost provides a valuable framework for understanding the challenges SMEs face when engaging in OI and our findings suggest that transaction costs related to negotiating agreements, managing risks, and protecting intellectual property are significant barriers for SMEs. As noted by Teece (1986), minimizing transaction costs is crucial for maintaining successful collaborations with external partners. The study also echoes the findings of Ahuja (2000), who emphasized the importance of reducing transaction costs in external collaborations to foster innovation. By managing transaction costs effectively, SMEs can improve the efficiency of their collaborations and reduce the risks associated with OI. The practical implication is that SMEs should carefully evaluate potential external collaborations, weighing the benefits against the costs involved. Additionally, SMEs should invest in mechanisms to protect their intellectual property and ensure that agreements are clear and enforceable to minimize the potential for opportunistic behaviour (Teece, 2007).

The role of appropriability regimes in safeguarding innovation investments is another critical finding of this study. An appropriability regime refers to the mechanisms organizations use to protect their intellectual property and prevent knowledge spillovers (Teece, 1986). Our findings suggest that SMEs must establish robust appropriability regimes to mitigate the risks associated with sharing proprietary knowledge with external partners. This is particularly important in OI, where the flow of knowledge between organizations can be difficult to control (Chesbrough, 2003). In line with the work of Arora and Gambardella (1990), our study highlights the need for SMEs to develop strategies for managing the risks of knowledge leakage and ensuring that innovation investments are protected. SMEs can implement formal agreements, such as non-disclosure agreements (NDAs) or intellectual property (IP) contracts, to safeguard their innovations and prevent misuse by external parties. The practical implication of this finding is that SMEs should not only focus on fostering external collaborations but also develop a clear and comprehensive strategy for protecting their intellectual assets in the OI process. SMEs must invest in creating a supportive organizational culture that fosters strategic external partnerships to manage transaction costs effectively and protect the intellectual property to succeed in OI. These efforts will help SMEs improve their innovation capabilities, enhance their competitiveness and drive long-term growth. The integration of OI practices into SMEs' strategic frameworks offers significant benefits, but it requires careful planning and commitment. By incorporating these findings, SMEs can build a more resilient and adaptive innovation ecosystem that is better equipped to meet the challenges of the rapidly evolving business environment.

Limitations of study

This research covers only a small portion of open innovation techniques, so case studies could provide more in-depth insights on specific topics and leadership traits. A cross-sectional analysis may not fully capture open innovation, as more time is needed for a deeper investigation. The study may not be generalizable to other industries, such as low-tech or service sectors. Given the quantitative approach focussing on objective measures, future research could use qualitative methods to explore subjective perspectives. The findings are specific to the manufacturing industry and may not apply to other sectors. As OI is new to Asia, further research in industries like the service sector or micro industries is needed, as well as a broader exploration of low-tech industries.

Suggestions for future study

Time is essential to explore more organizations for a complete understanding of OI, and comprehensive studies require longitudinal approaches. Future research should consider the role of an appropriability regime as a mediating factor when comparing SMEs' success in implementing OI. Since this study was conducted in Singapore, investigating this link in both developing and developed countries would help generalize the findings. Given the quantitative focus on measurable metrics, future studies could adopt a qualitative approach to explore subjective factors. This study focussed on SMEs, but it would be interesting to compare OI in the government sector, which is less profit driven. Future research could explore the success of OI initiatives in SMEs aimed at sustainability and identify key factors for improving innovation efficiency. It would also be valuable to assess the potential benefits and drawbacks of OI in the context of business sustainability, focussing on the mechanisms that influence sustainability performance.

Conclusion

The key conclusions of this study indicate that factors such as integrative culture, management links, organizational citizenship behaviour and transaction costs influence the adoption of OI. The relationship between management ties and transaction costs is moderated by the appropriability regime. However, integrative culture and organizational citizenship behaviour were not directly associated with OI adoption. To foster OI, SMEs should prioritize strategic planning, encourage organizational citizenship behaviour through employee training, and develop an integrative culture where ideas flow freely across the organization. Building external relationships and reducing transaction costs will also enhance collaboration. The appropriability regime plays a critical role in mitigating risks and accelerating results and therefore in order to be successful, SMEs should engage in OI concept while addressing potential risks. Innovation policy is crucial for encouraging collaboration between SMEs and larger corporations to create new markets and business opportunities. By combining expertise and new business models, SMEs can innovate and expand into new markets and innovation policies should support and strengthen these efforts.

Appendix 1

see Tables 15, 16, 17, 18, 19, 20.

Table 15 Measurement model for 1st order—OCB

OCB	Items	Factor loading	Cronbach alpha	Composite reliability	AVE
Altruism	Altr1	0.841	0.798	0.861	0.756
	Altr2	0.897			
Conscientiousness	Consc1	0.913	0.847	0.909	0.770
	Consc2	0.764			
	Consc3	0.946			
Courtesy	Court1	0.914	0.847	0.908	0.768
	Court2	0.900			
	Court3	0.812			
Civic Virtue	Civic2	0.902	0.911	0.944	0.848
	Civic3	0.935			
	Civic4	0.926			
Sportsmanship	Sport1	0.898	0.812	0.889	0.728
	Sport2	0.826			
	Sport3	0.834			

Table 16 Measurement model for 1st order—IC

OC	Items	Factor loading	Cronbach alpha	Composite reliability	AVE
Employee Development	Emdev2	0.947	0.868	0.938	0.883
	Emdev3	0.932			
Harmony	Harm1	0.794	0.887	0.923	0.750
	Harm2	0.822			
	Harm3	0.948			
	Harm4	0.892			
Customer Orientation	Cusor2	0.960	0.917	0.960	0.923
	Cusor3	0.962			
Innovative culture	Incul1	0.877	0.876	0.915	0.729
	Incul2	0.755			
	Incul3	0.899			
	Incul4	0.878			

Table 17 Measurement model for 1st Order—MT

MT	Items	Factor loading	Cronbach alpha	Composite reliability	AVE
Ties with Managers	Matie1	0.854	0.805	0.885	0.719
	Matie2	0.822			
	Matie3	0.868			
Ties with Research Centres	Matie4	0.904	0.889	0.931	0.819
	Matie5	0.912			
	Matie6	0.898			
Ties with Officials	Matie7	0.953	0.960	0.974	0.926
	Matie8	0.973			
	Matie9	0.961			

Table 18 Measurement model for 1st order—TC

TC	Items	Factor loading	Cronbach alpha	Composite reliability	AVE
Asset specificity	Asset1	0.877	0.785	0.875	0.701
	Asset2	0.769			
	Asset3	0.861			
Environment uncertainly	Envon1	0.883	0.829	0.896	0.743
	Envon2	0.932			
	Envon3	0.762			
Behavioural uncertainly	Behunc1	0.899	0.758	0.852	0.742
	Behunc2	0.822			
Technology competency	Tech1	0.923	0.846	0.929	0.867
	Tech2	0.938			
Degree of competition	Decom1	0.826	0.707	0.828	0.707
	Decom2	0.856			

Table 19 Measurement model for 1st order—AR

Appropriability regimes	Items	Factor loading	Cronbach alpha	Composite reliability	AVE
	AR3	0.879	0.882	0.932	0.873
	AR4	0.987			

Table 20 Measurement model for 1st order—OI

Open innovation	Items	Factor loading	Cronbach alpha	Composite reliability	AVE
	OI1	0.714	0.705	0.775	0.535
	OI2	0.763			
	OI3	0.716			

Abbreviations

SMEs	Small medium enterprises
OI	Open Innovation
OCB/s	Organizational citizenship behaviours
IC	Integrative culture
MT	Managerial ties
TC	Transactional cost
AR	Appropriability regime

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Declarations

Ethics approval and consent to participate
We testify on behalf of all co-authors that our article titled: Driving Open Innovation in SMEs: The Role of Organizational and Strategic Dynamics has not been published in whole or in part elsewhere. The manuscript is not currently being considered for publication in another journal. All the authors have been personally and actively involved in substantive work leading to the manuscript and will hold themselves jointly and individually responsible for its content. Date: 26/6/2024.
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